

Lecture 35 (April 30)

Colorings

Definition (k -colorings).

Let $G = (V, E)$ be a graph. A proper **k -coloring** of G is a function $c: V \rightarrow \{1, 2, \dots, k\}$ such that for all edges $(u, v) \in E$, $c(u) \neq c(v)$.

(The values $1, 2, \dots, k$ represent colors.)

For example, consider $G = K_3$. We can see that there are no 1-colorings or 2-colorings (only graphs with no edges are 1-colorable, while only bipartite graphs are 2-colorable).

There are six 3-colorings (where you give every vertex a different color) and $4 \cdot 3 \cdot 2 = 24$ 4-colorings.

Theorem 1 (Chromatic polynomial).

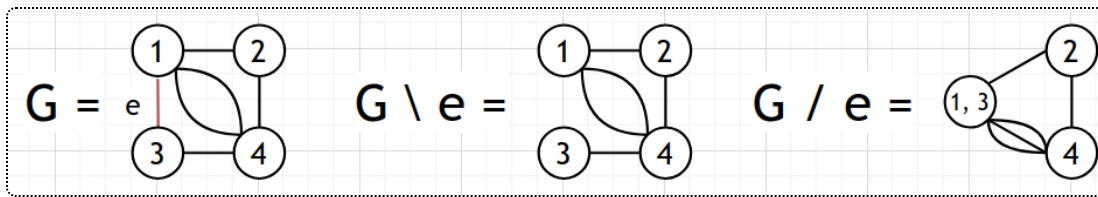
Given a graph G , there exists a unique **chromatic polynomial** $\chi_G(t)$ such that for any nonnegative integer k , the number of k -colorings of G equals $\chi_G(k)$.

Moreover, this polynomial has integer coefficients.

Proof. We induct on $|E|$.

Base case: $E = \emptyset$, any function is a coloring, and $\chi_G(k) = k^{|V|}$.

Inductive step: We use deletion-contraction. Assume that $|E| \geq 1$; pick any edge $e = (u, v)$. We consider the graph $G \setminus e$ where it's deleted and G / e where it's contracted:



Note that even if the original graph is simple, we will need to allow multiple edges since contraction may cause that. In this case, it doesn't matter since for colorings, having multiple edges is the same as having one edge (it just means that the two vertices are of different colors.)

Lemma 2 (Deletion-contraction on colorings).

$$\chi_G(k) = \chi_{G \setminus e}(k) - \chi_{G/e}(k).$$

Proof of Lemma. The deleted graph must have more colorings since it has fewer restrictions. Let's consider any coloring c of $G \setminus e$; there are two cases.

1. $c(u) \neq c(v)$. Then c is a k -coloring of G .
2. $c(u) = c(v)$. Then contracting yields a k -coloring of G / e .

We can note that the reverse operations also work, so indeed $G \setminus e$ has as many k -colorings of G and G / e combined.

Since both $G \setminus e$ and G / e have one fewer edge than G , by the inductive hypothesis, both $\chi_{G \setminus e}(t)$ and $\chi_{G/e}(t)$ are polynomials, and $\chi_G(t)$ (their difference) is too.

We can also note that χ_G has integer coefficients and is monic with degree $n = |V|$ (since the only way to get a monomial of degree n is to delete all edges).

Here are some examples:

- $\chi_{K_n}(t) = t(t-1)(t-2) \dots (t-n+1)$

- For any tree on T vertices, $\chi_T(t) = t(t-1)^{n-1}$ (start with any vertex and keep adding leaves).

Acyclic orientations

Definition (Acyclic orientations).

An **acyclic orientation** of an undirected graph G is a way to orient all edges such that there are no directed cycles.

$\text{AO}(G)$ is the number of acyclic orientations of G .

For example, $\text{AO}(K_3) = 2^3 - 2 = 6$. In general, $\text{AO}(n\text{-cycle}) = 2^n - 2$.

For any forest F (i.e. a graph without cycles), $\text{AO}(F) = 2^{|E|}$. Every acyclic orientation of K_n must come from ordering every vertex and orienting each edge towards the larger one, so $\text{AO}(K_n) = n!$.

Note that having multiple edges again does not affect $\text{AO}(G)$, since all edges between the same vertices must be oriented the same way to avoid 2-cycles.

Theorem 3 (Richard Stanley (1973)).

For a graph G with n vertices,

$$\text{AO}(G) = (-1)^n \chi_G(-1).$$

Proof. Once again, we use induction on $|E|$ and deletion-contraction.

Base case. $|E| = 0$; we have $1 = (-1)^n (-1)^n$.

Inductive step. We prove the following lemma.

Lemma 4 (Deletion-contraction for $\text{AO}(G)$).

$$\text{AO}(G) = \text{AO}(G \setminus e) + \text{AO}(G / e).$$

Proof of Lemma. Let $e = (u, v) \in E$ and $\mathcal{O}$ be an acyclic orientation of $G \setminus e$. We will consider two cases that describe how e can be oriented:

1. There exists a directed path in $\mathcal{O}$ between u and v (either $u \rightarrow v$ or $v \rightarrow u$).
Note that there cannot be a directed path in both directions, since that would require a cycle. In this case, there is exactly one valid way to orient e (in the same direction as the path).
2. If there are no directed paths between u and v , either orientation is fine.

If we now contract e in $\mathcal{O}$, we end up with a valid acyclic orientation of G / e only in case 2 (case 1 yields a directed cycle.) Thus in case 1, $\mathcal{O}$ contributes 1 to both sides of the lemma, and in case 2, $\mathcal{O}$ contributes 2 to each side, so indeed they are equal.

Now, with the inductive hypothesis, we get

$$\begin{aligned}
 \text{AO}(G) &= \text{AO}(G \setminus e) + \text{AO}(G / e) \\
 &= (-1)^n \chi_{G \setminus e}(-1) + (-1)^{n-1} \chi_{G/e}(-1) \\
 &= (-1)^n (\chi_{G \setminus e}(-1) - \chi_{G/e}(-1)) \\
 &= (-1)^n \chi_G(-1),
 \end{aligned}$$

as desired.

Exercise (future problem set problem): Compute the number of acyclic orientations of a complete bipartite graph.

Lecture 36 (May 2)

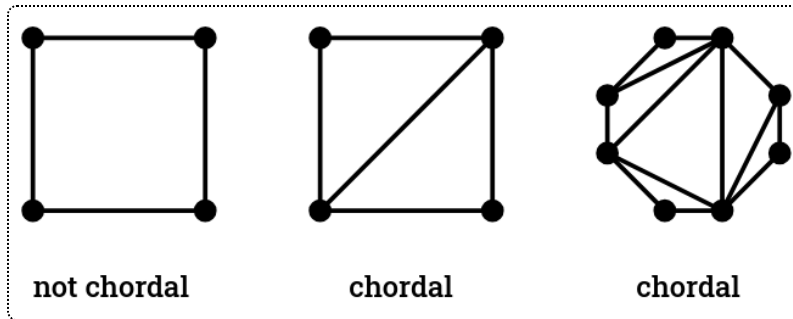
Chordal graphs and perfect elimination orderings

Definition (Chordal graphs).

A graph G is **chordal** if every cycle $C = (u_1, u_2, \dots, u_m, u_1)$ with $m \geq 4$ vertices has a chord, i.e. an edge between u_i and u_j where

$$\{i, j\} \notin \{\{1, 2\}, \{2, 3\}, \dots, \{m, 1\}\}.$$

For example, if we start off with a graph with only one cycle, we essentially need to triangulate it to make it chordal.



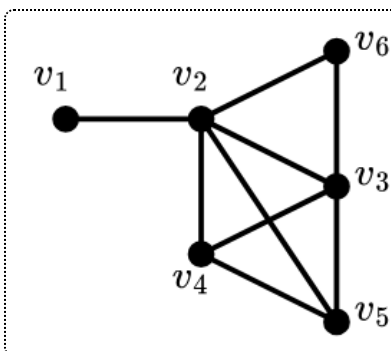
Definition (Perfect elimination ordering).

A **perfect elimination ordering** of G is a way to order its vertices $v_1, v_2, \dots, v_n$ such that for any $i = 2, 3, \dots, n$, the set

$$\{v_j \mid j < i \text{ and } v_j \text{ is adjacent to } v_i\}$$

is a clique in G (i.e. a complete subgraph.)

Essentially, if we construct the graph by sequentially adding vertices, each time we connect a new vertex to a clique of existing vertices. Note that the perfect elimination ordering may not be unique.



Theorem 5 (Fulkerson-Gross (1965)).

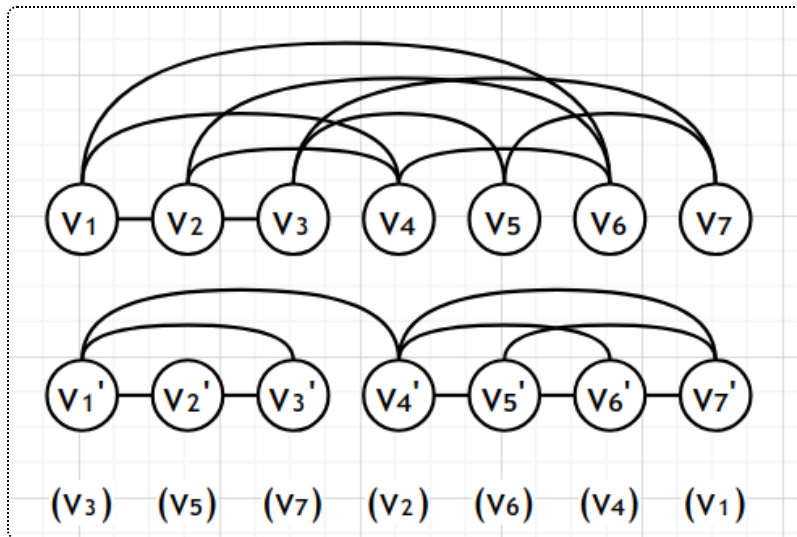
G is chordal if and only if G has a perfect elimination ordering.

The "only if" direction is easy: consider any cycle in G . Let v_i be its maximal vertex in a perfect elimination ordering. Then it's connected to the two adjacent

vertices in the cycle (call them v_j and v_k). Since i is maximal, $j, k < i$, and as such v_j and v_k must be part of the same clique, so (v_j, v_k) is a chord as desired.

Let $v_1, \dots, v_n$ be a perfect elimination ordering of G . Define a_i to be the number of vertices v_j where $j < i$ and (v_i, v_j) is an edge of G (i.e. the size of the clique that i was attached to).

For example, here are two different elimination orderings of the same graph:



Here are the corresponding a -values:

i	1	2	3	4	5	6	7
a_i	0	1	1	2	1	3	2
a'_i	0	1	2	1	1	2	3

We notice that the values a'_i are exactly a rearrangement of the values a_i . This isn't a coincidence; in fact we have the following theorem:

Theorem 6 (Perfect elimination orderings).

$$\chi_G(t) = \prod_{i=1}^n (t - a_i)$$

(Sidenote: $\chi_G(t)$ is also sometimes known as the "characteristic polynomial"; it also has significance in e.g. hyperplane arrangements which can be thought of as a multidimensional analogue.)

The idea is simple. Imagine coloring the vertices in the order of the perfect elimination ordering; each vertex v_i will have $(t - a_i)$ different options (the a_i vertices it's connected to form a clique, so they each eliminate one distinct color.)

Tutte polynomial

There is a nice graphical invariant to describe deletion-contraction recurrences, the **Tutte polynomial** (aka the Tutte-Whitney polynomial or the dichromate of a graph). This can be thought of as a generalization of the chromatic polynomial (which is also defined with deletion-contraction).

Theorem 7 (Tutte (1954)).

For any graph $G = (V, E)$ (potentially with multiple edges and loops), there is a unique polynomial $T_G(x, y)$ that satisfies:

1. $T_G = T_{G \setminus e} + T_{G/e}$ for any edge e which is not a loop nor a bridge.
2. For any bridge e , $T_G = x \cdot T_{G/e}$
3. For any loop e , $T_G = y \cdot T_{G \setminus e}$
4. If G has no edges, $T_G = 1$.

Recall that a bridge or isthmus is an edge that, if removed, increases the number of connected components in the graph, and a loop is an edge between a vertex and itself.

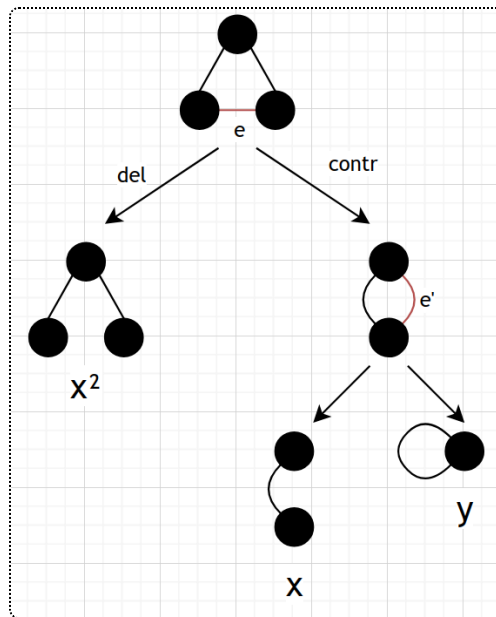
Corollary 8 (Base case).

If G has i bridges, j loops, and no other edges,

$$T_G(x, y) = x^i y^j.$$

(i.e. G has no cycles of length ≥ 1).

To find the Tutte polynomial of any graph, it is enough to use case 1 from the definition and this base case. For example, $T_{K_3}(x, y) = x^2 + x + y$:



It is not obvious that this polynomial actually exists (i.e. that any way to do deletion-contraction yields the same result T_G). So, it would be nice to find a non-recursive formula for T_G , and there are actually several of them, like the following:

Definition (Whitney's corank-nullity polynomial of G).

We define $W_G(x, y)$ as such:

$$W_G(x, y) = \sum_{A \subseteq E} x^{k(A) - k(E)} \cdot y^{k(A) + |A| - |V|}.$$

where $k(A)$ is the number of connected components of the subgraph $H = (V, A)$.

It's called the "corank-nullity" polynomial since it can be represented in terms of the oriented incidence matrix of the graph (the exponents are the corank and the nullity).

Theorem 9 (Tutte and Whitney polynomials).

$$T_G(x, y) = W_G(x - 1, y - 1).$$

Proof. Verify the deletion-contraction recurrences 1 through 4 for $W_G(x-1, y-1)$. This immediately proves Tutte's theorem.

Note that all graphical invariants mentioned before (e.g. number of spanning trees, number of acyclic orientations, number of connected subgraphs), are all specializations of the Tutte polynomial (i.e. they can be obtained by plugging in some values for x and y). The tree inversion/parking function polynomial is also a special case of T_{K_n} .

Lecture 37 (May 5)

We will continue discussing the Tutte polynomial. Note that the recursive formula implies that all of its coefficients are positive integers.

Assume that a graph G is connected. (Note that if G comprises multiple connected components, its Tutte polynomial is just the product of those of the components.)

Then $T_G(1, 1) = t(G)$ (the number of spanning trees of G). By considering statistics on these spanning trees, we can come up with a third formulation of T_G :

Tutte's internal/external activities

Fix a total ordering of edges $E = \{e_1 < e_2 < \dots\}$ in a connected graph G . Let $T \subset G$ be a spanning tree. We want to discuss "internal" and "external activities" of these trees.

Definition (Internal activity).

Let $e \in T$ be an edge. Removing e yields two subtrees T_1 and T_2 , though there may be some edges in $G \setminus T$ that connect them. We call e **internally active** if it's the smallest edge connecting T_1 and T_2 .

We define $\text{int}(T)$ (the **internal activity** of T) to be the number of internally active edges.

In other words, if you can replace e with a smaller edge e' and maintain that $(T \setminus \{e\}) \cup \{e'\}$ is a tree, e is not internally active.

Definition (External activity).

Consider an edge $e \in G \setminus T$. $T \cup \{e\}$ contains a unique cycle C , and e is **externally active** if it's the smallest edge in C . $\text{ext}(T)$, the **external activity**, is the number of such edges.

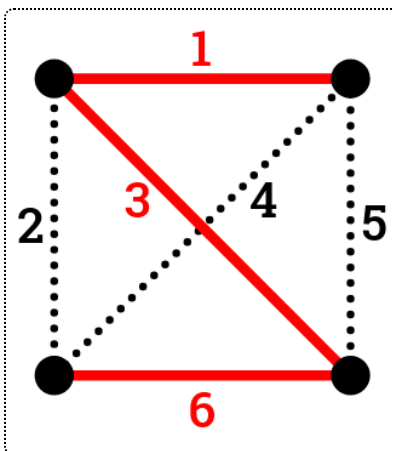
This can be thought of as a "dual" of internal activity, since it considers edges outside the tree.

Theorem 10 (Tutte).

$$T_G(x, y) = \sum_{\substack{T \text{ spanning} \\ \text{tree of } G}} x^{\text{int}(T)} \cdot y^{\text{ext}(T)}.$$

In particular, even though $\text{int}(T)$ and $\text{ext}(T)$ depend on the specific edge ordering, the polynomial does not.

For example, in this labeling of K_4 where T comprises edges 1, 3, and 6, the only internally active edge is 1 and the only externally active edge is 2, so $\text{int}(T) = \text{ext}(T) = 1$.



In practice, there is no simple general computation for the Tutte polynomial (like the matrix-tree theorem); they will all involve at least summing over all spanning trees.

Proof idea. Show that the right hand side satisfies the deletion-contraction recurrence $T_G = T_{G \setminus e} + T_{G/e}$ specifically where e is the maximal edge. Since T_G is a unique polynomial, this is enough.

Specializations of the Tutte polynomial

As mentioned before, every single deletion-contraction relation can be written as a special form of the Tutte polynomial:

Theorem 11 (Special cases of T_G).

- $T_G(1, 1)$ is the number of spanning trees of G
- $T_G(2, 1)$ is the number of forests in G
- $T_G(1, 2)$ is the number of connected subgraphs $(V, E' \subseteq E)$ in G
- $T_G(2, 0)$ is the number of acyclic orientations in G
- $T_G(0, 2)$ is the number of "strongly connected orientations" in G
- $T_G(2, 2) = 2^{|E|}$

(A strongly connected orientation is a way to orient all edges of G so that for any two vertices v, w , there is a path $v \rightarrow w$ and a path $w \rightarrow v$.)

Note that we consider $0^0 = 1$ when setting $x = 0$ or $y = 0$ and using e.g. the internal/external activity definition of T_G (that way it doesn't just add to 0).

Proof idea. Just verify the deletion-contraction properties and base cases of the right hand side.

Theorem 12 (Chromatic polynomial).

$$\chi_G(t) = (-1)^{|V| - k(G)} t^{k(G)} \cdot T_G(1 - t, 0).$$

Recall that $k(G)$ represents the number of connected components of G .

This one can be proved the same way.

Here's another one that we'll discuss further next time:

Definition (Flow polynomial).

The **flow polynomial** $C_G(t)$ is the number of *nowhere-zero t -flows* on G .

A nowhere-zero k -flow is an assignment of values in $\{1, 2, \dots, k-1\}$ to an oriented version of G so that each vertex has net-zero flow mod k .

Theorem 13 (Flow polynomial and Tutte polynomial).

$$C_G(t) = (-1)^{|E|-|V|+k(G)} \cdot T_G(0, 1-t).$$

Also, an interesting fact about planar graph duals:

Theorem 14 (Planar duals).

If G is a planar graph drawn in the plane and G^* is its planar dual, we have

$$T_G(x, y) = T_{G^*}(y, x).$$

One more theorem for today, that follows directly from the external activity formula:

Theorem 15 (Parking and tree-inversion polynomials).

$$T_{K_{n+1}}(1, y) = p_n(y) = I_n(y)$$

(Recall that $p_n(y)$ is the parking polynomial and $I_n(y)$ is the tree-inversion polynomial.)

In fact, if we lexicographically order the edges of K_{n+1} , externally active edges correspond exactly to tree inversions.

Note that this, together with $T_G(1, 2)$, lets us compute the number of connected subgraphs on a complete graph (i.e. problem 1 on the problem set). However, if you want to use this, you will have to prove everything you use, so it is

recommended to use a different recurrence and only use this theorem to check your answer.